Practice Quiz: Learning / Clinical Translation (Research Methods)

Name: __ ***Date:*** __

Part A: Multiple Choice

1. Computational psychiatry characterizes mental disorders by: A) DSM categories B) Dysfunction in specific computational variables — precision, model structure, preferences, and belief updating — rather than descriptive symptom checklists C) Brain volume D) Genetic markers alone

2. The Active Inference account of hallucinations in schizophrenia proposes: A) Random brain activity B) Aberrant sensory precision — overly precise sensory priors generate percepts without external stimulation C) Intentional fantasy D) Attention deficit

3. Anhedonia in depression is modeled as: A) Loss of senses B) Reduced precision on positive (reward) prediction errors — leading to failure to learn from and be motivated by positive outcomes C) Increased appetite D) Sleep disturbance

4. The translational pipeline includes: A) Only theory B) Five steps — computational theory, behavioral validation, neural validation, pharmacological validation, and treatment development C) Only drug testing D) Only symptom description

5. Digital phenotyping enables: A) One-time assessment B) Longitudinal real-time tracking of computational parameters using smartphone tasks — potentially predicting relapse before clinical symptoms emerge C) Only surveys D) Brain imaging at home

6. Computational enrichment in clinical trials: A) Enrolls all patients B) Selects patients whose computational phenotype matches the treatment's proposed mechanism — improving statistical power and clinical relevance C) Excludes all modeling D) Uses only placebo

7. Interoceptive inference dysfunction in anxiety is characterized by: A) Reduced bodily awareness B) Excessively precise interoceptive priors that generate persistent alarm signals —

the body is predicted to be in danger even when physiologically safe, driving anxious affect and avoidance behavior C) Normal precision weighting D) Only cognitive symptoms

8. Pharmacological validation in computational psychiatry means: A) Testing any drug effect B) Administering drugs that selectively modulate specific computational variables (e.g., dopamine agonists for precision, ketamine for NMDA-mediated prediction errors) to test whether the predicted behavioral changes occur C) Only using placebos D) Skipping behavioral testing

Part B: Short Answer

1. A patient with treatment-resistant depression shows abnormally low reward prediction error precision ($\zeta\_reward = 0.12$, healthy mean = 0.85). Their clinician wants to use this to select a treatment. Explain how this computational phenotype could guide treatment selection between SSRIs, dopamine agonists, and behavioral activation therapy. (200 words)

2. Digital phenotyping collects smartphone data (GPS, typing patterns, social media activity) to track computational parameters longitudinally. Describe one specific computational parameter that could be tracked this way and explain what pattern would signal impending relapse in a patient with bipolar disorder. (200 words)

Part C: Essay Questions

1. Choose a psychiatric disorder not covered in the module (e.g., OCD, PTSD, ADHD, addiction). Develop a full Active Inference computational model: (a) identify the primary computational dysfunction (precision, structure, preferences, or updating), (b) specify the generative model parameters, (c) derive behavioral predictions, (d) design a behavioral task, (e) propose a treatment targeting the computational dysfunction. (600 words)

2. Write a grant proposal abstract (300 words) for a computational psychiatry study using Active Inference. Include: scientific rationale, specific aims, methods (task, model, analysis), clinical significance, and innovation.

3. Critically evaluate the ethics of computational phenotyping in psychiatry. Address: (a) privacy of computational data, (b) risk of algorithmic bias in diagnosis, (c) informed consent when patients may not understand computational models, (d) the danger of reducing mental illness to parameter values, (e) the potential benefit of more precise treatments. (500 words)